International Bachelor Mechanical, Materials and Aerospace Engineering
Transversal Project Raport

Landing Gear

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1 Abstract

Landing gear, being one of the most fundamental parts of an aircraft, is a complex structure that plays a vital role in takeoff, taxiing, landing, and all ground operations. This project highlights the conceptualization, design and construction process of a fully mechanical model of a main landing gear, focusing on its self-locking mechanisms, its retraction system and its load-bearing truss design. In our sized-down model, we have incorporated a manually-controlled actuator for the retraction and extension of our retractable main landing gear model. All electrical, hydraulic, and pneumatic components have been replaced with an equivalent mechanical components to be able to better manipulate our model and meet the project requirements.

We will also delve into the different designs of landing gears, the importance of them and the significance of our mechanical model compared to a real-life, full-size landing gear, specifically the Boeing 737 model. A kinematic and part-specific breakdown of the model is also discussed.

2 Introduction

Airplanes predominantly operate in the air. However, when they are on the ground, they require a mechanism that will manage their weight and impact on the ground. This is mainly what a landing gear is designed for. The primary purpose of a landing gear is to aid the plane during taxiing, landing, take off and any ground operations. It is also essential to absorb and disperse the kinetic energy caused by landing. The deployment and retraction of the landing gear are controlled by hydraulic or electric actuators, which enable its extension during takeoff and landing, as well as its retraction during flight. Additionally, shock absorbers integrated into the landing gear system help to cushion landings, thereby minimizing stress on the airframe and improving passenger comfort.

2.1 Evolution of landing gear

During the early years of aviation, pioneers in aeronautics faced a lot of challenges to create a structure that would support the weight of the airplane and help it to land safely on the ground.

Remarkably, the first-ever flight designs relied on the human body- the pilot's legs- to initiate take off and brake upon landing.

In 1903, the Wright Brothers fitted their model with skis that would allow the plane to land on the ground. This was considered the forerunner of the coming models of landing gears. [11]

In 1906, the Santos Dumont brothers equipped their 14-bis with small wheels that were devoid of shock absorbers, which resulted in rather jarring landings. [11] The rudimentary suspension system, which was either in its early stages or entirely absent, did not necessitate significant specifications due to the limited speed and weight of the initial aircraft, constructed from wood and canvas. Consequently, it comprised a straightforward design utilizing rubber bands and lattice frameworks. Moreover, as mentioned, the size of the wheels on the earliest prototypes were considerably small. As a matter of fact, most of these wheels were originally designed to be used for bicycles.

During that period, however, landing gears were fixed since they were still flying at slower speeds and did not have the sizes or weights that we have for newer aircrafts.

It wasn't until 1920 with the introduction of the Dayton Wright RB-1 in which the world saw the emergence of the first ever retractable landing gear which pilots manipulated manually using a crank-which is similar to the design that will be studied all throughout this paper. [11] [3]

Fixed landing gear is still found on small aircrafts to this day -which reduces aerodynamic resistance- since the weight of a fairing is significantly smaller than the weight of the system of a retractable landing gear.

The important increase in use of aluminum on aircrafts led to not only the increase in performance but also the weight of the aircraft. This means that more complicated structures of the landing gear had to come into play to be able to support the heavier impact of the airplane upon landing and absorb more shock caused by this gradual increase in the weight and speed of the aircrafts. That led to the evolution of the braking system and the steering system in turn which had to be designed for these heavier weights and higher speeds.

Lastly, one of the most pivotal eras for aviation that is worth mentioning is World War II, in which we saw the birth of some of the best inventions in the aeronautical field that are still being used to this day and incorporated in some of the newest models of aviation.

2.2 Types of Landing Gears

Different landing gears exist and were created based on the size and purpose of the plane. These different types can vary based on design or configuration. [9]

2.2.1 Types of landing gears based on design

The two primary designs of landing gear are fixed or retractable. You can see, for reference, Figure 1a and Figure 1b which correspond to a fixed and retractable landing gear design respectively.



(a)



(b)

Figure 1: Possible problems caused by incorrect truss dimensions.

A fixed landing gear, as seen in Figure 1a, is always in the extended position. It is usually used on smaller aircrafts, trainers and certain utility planes due to its robust and reliable design. It is also made up of less parts which means it is easier to maintain and less costly overall. Nonetheless, having the landing gear always extended causes significant increase in aerodynamic drag which in turn leads to more fuel consumption. [9]

A retractable landing gear, as seen in Figure 1b, is the design that folds into the fuselage and the wing for the nose and main gear respectively. It is a design that has allowed planes to have less aerodynamic drag, making them more fuel efficient and faster. On the other hand, this design is more complex and requires more complex parts such as the hydraulic and mechanical system which leads to it being more costly, more difficult to maintain and prone to a higher number of failures whether it be in terms of use (pilot error) or in terms of it being poorly engineered. [9]

2.2.2 Types of Landing Gears Based on Configuration

There are two primary types of landing gear configurations: conventional (Figure 2a) and tricycle (Figure 2b) landing gear, which can be identified based on their structural arrangement. The configuration of a conventional landing gear, as shown in Figure 2a, was the earliest employed, primarily due to the arrangement of the wings and engines, as it was most compatible with the center of gravity of early aircraft designs. Conventional landing gear, also known as taildragger,typically consists of two main struts positioned beneath the wings or at a similar height, along with a wheel located at the tail. [9]

While this design offers aerodynamic advantages, particularly in the case of fixed and robust retractable landing gear, it also presents several drawbacks. These include limited ground



(a) (b) (c)

Figure 2: Possible problems caused by incorrect truss dimensions.

visibility, challenges related to lift during takeoff due to the aircraft's inclined stance, and difficulties in landing under turbulent conditions or crosswinds.[9]

On the other hand, the tricycle landing gear configuration, as shown in Figure 2b features two main struts located beneath the wings or fuselage, complemented by an additional strut positioned under the nose, which is equipped with a directional control mechanism.

The tricycle landing gear significantly simplifies the landing process, even in the presence of crosswinds or tailwinds. Furthermore, it mitigates the risk of the aircraft overturning, facilitating a horizontal touchdown. [9] There is a specialized configuration that is also commonly used known as the ski landing gear as shown in Figure 2c.

The purpose of ski landing gear is to enable aircraft with skis instead of wheels to land smoothly on frozen or snow-covered surfaces. Ski gear is indispensable for scientific missions, emergency services, and other operations. The skis are frequently retractable or interchangeable with wheels, allowing pilots to operate on both snow and regular runways. Ski landing gear limits the aircraft's versatility by limiting the aircraft's operations to snow-covered surfaces. Landing on icy surfaces with ski gear also requires careful control because reduced friction can affect steering and braking.

2.3 Landing Gear Configurations

Landing gear is an extremely complex configuration of many different systems and subsystems supporting their functioning. There are multiple minute parts in a landing gear that we won't be able to list and talk about all of them, in this section of the report we will only talk about the most noticeable parts and especially the retraction system and locking mechanism since our project mostly revolves around creating a working retraction and locking system.

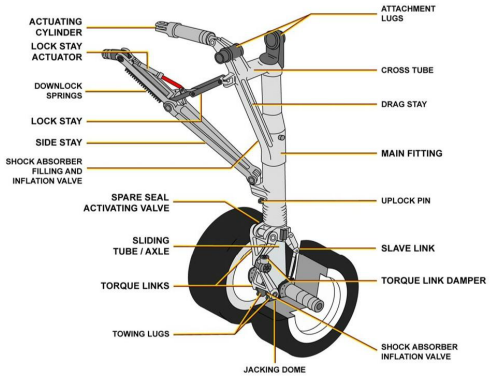
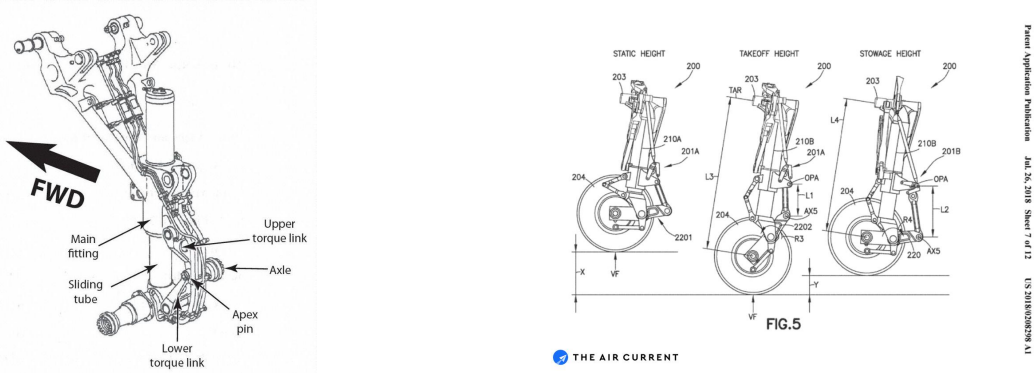


Figure 3: Main components of landing gear.

2.3.1 Shock Strut

The shock strut assembly or in our project which is called the "main shaft", is a very crucial part to a landing gear. This assembly holds the aircraft's weight while on the ground, also maintaining stability and sometimes providing steering. This assembly also absorbs impact with a suspension system, this prevents excessive shock going to the fuselage and damage to other parts of the aircraft. There are 5 main components in a shock strut: Outer cylinder, Inner cylinder, Metering pin, Hydraulic fluids and Nitrogen chamber (suspension), Torque links and shimmy dampers.



(a) **(b)**

Figure 4: Suspension [8].

When the aircraft touches down the runway, the inner cylinder is pressed to the outer cylinder which forces the hydraulic fluids through the metering pin. The hydraulics then resists the compression by transferring the impact force into heat, nitrogen is also compressed which furthermore absorbs the impact of the landing. Once all the landing force is absorbed, nitrogen expands to push the strut back to its original placement.

2.3.2 Retraction System

This system is responsible for raising the landing gear during flight and lowering it during landing, it uses a hydraulic system to complete the task of lowering and raising the gear. Apart from the hydraulics, it is a very well put and clever organization of the parts and wheels in order to fit the landing gear inside the wing. One of our main tasks during the project was to create a working retraction system while not using hydraulic systems.

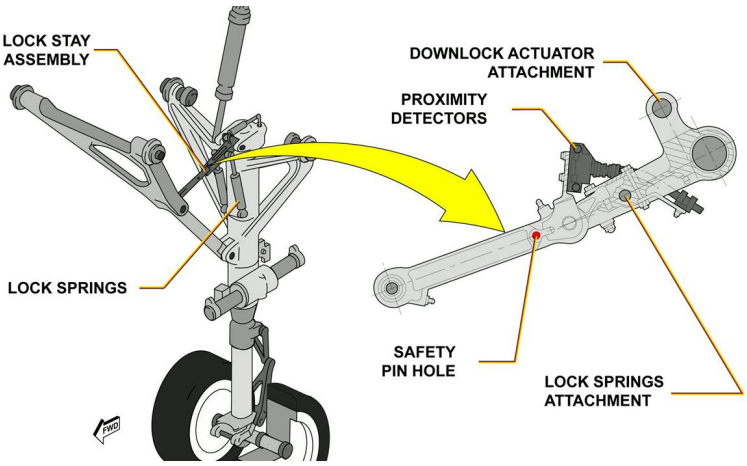


Figure 5: Retraction system [7].

Here we will have the main parts of this system. First we will have the hydraulic actuator which acts as the main source of power for moving the landing gear, this is also our main focus of the project since we have to design an actuator using only mechanical forces. The locking mechanism is also one of our focus, since we have to design a lock that both works while the gear is retracted and extended, normally this uplock and downlock mechanism is fitted with a mechanical hook and latch system, the lock's job is to keep the gear from being moved out of place from turbulence or vibrations. For the safety measures we will have an emergency manual extension system, which is used via a lever which the pilot will pull to extend the landing gear using gravity, this system is used when the hydraulic actuator fails.

2.3.3 Locking System

The locking system is one of the main systems is the retraction system, the lock ensures that the landing gear stays perfectly in place, there are two main locks for two parts of the flight. Downlock, this lock is used when the plane is landing or is on the ground, this lock keeps the gear undisturbed by outside factors such as turbulence and aerodynamic forces.

When lowering down the landing gear, the actuator is extended to lower the gear using hydraulic fluids, once the gear fully extended, a spring-load lock mechanism ensures that the gear stays in place even when the hydraulic actuator doesn't work. For retracting the landing gear, the actuator is retracted which in turn retracts the landing gear simultaneously, after the landing gear is fully retracted, an uplock hook is engaged to secure the gear and just as the downlock system, the uplock system does not require hydraulic actuator to stay locked. During these operations there will also be a sensor that feeds information to the pilot on which the gear is fully retracted/extended or not.

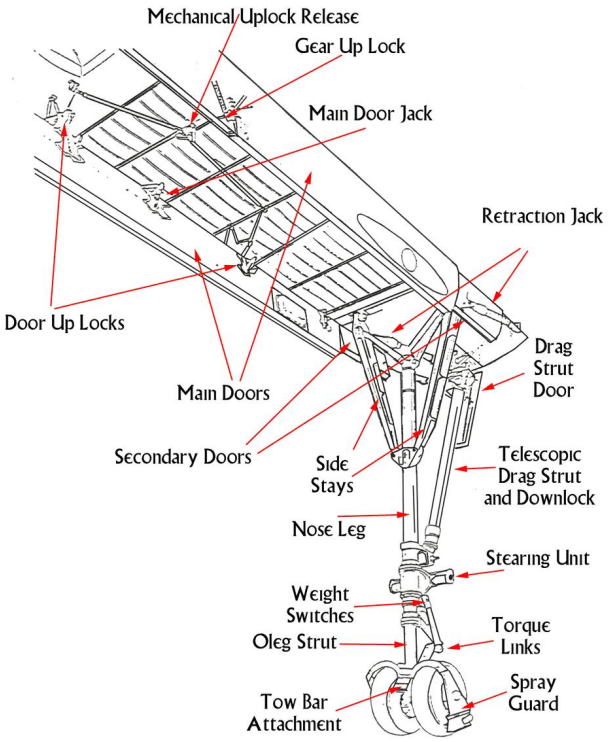


Figure 6: Locking system [2].

3 Model

3.1 Design

The main landing gear, which was the model that we designed, is known as the backbone of the aircraft's landing system. It is intended to support the majority of the aircraft's weight and is situated beneath the wings or fuselage.

As the main mechanism for sustaining the aircraft's weight during landing and ground operations, the main landing gear is a fundamental part of an aircraft's landing system. This crucial component, which is positioned below the wings or fuselage of the aircraft, is essential to guaranteeing secure and effective landing operations.

This is why we decided to study the main landing gear and get inspired from it for the creation of our all-mechanical smaller scale model.

For our design of the main landing gear component, we had to mainly focus on the design of the trusses similar to the ones used on the Boeing 737. If we were to simplify that into the most straightforward and efficient form possible, we would get the following trusses

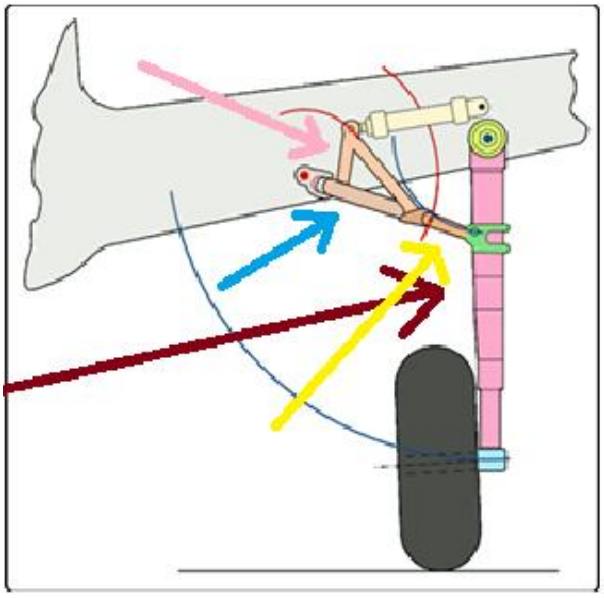


Figure 7: Wheel

- The **wheel beam**, which is normally known as the main fitting. On a plane, that would be the part that has the shock absorber filling and inflation valve.
- The **truss #1** of our truss was the self-locking aid for when the truss is fully extended. This is known as the side stay on a regular plane's landing gear.
- The **truss #2** of our truss is the section which connects the side stay to the fuselage while also allowing it to rotate when we need it to go back into retraction position. On a regular plane, this is the section that normally has the downlock springs.
- The last, and probably the most important part of our model, when it comes to actuator, was the **triangle truss** section, which was the connection between the actuator and the rest of the truss

As a whole, the design was made to handle the loads it is subjected to, perform a retraction mechanism, and create and downward and upward locking system (during retraction and extension modes).

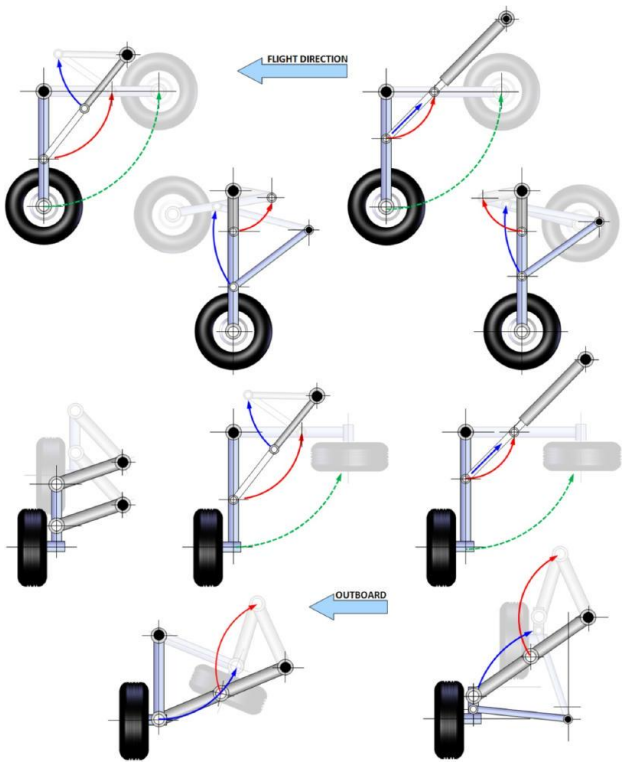


Figure 8: Diagrams of different designs for the trusses of a retractable landing gear [5].

We opted to go with the design shown on Figure 9, since the triangle allows the force to be exerted with an angle close to 90 degrees for most of the motion.

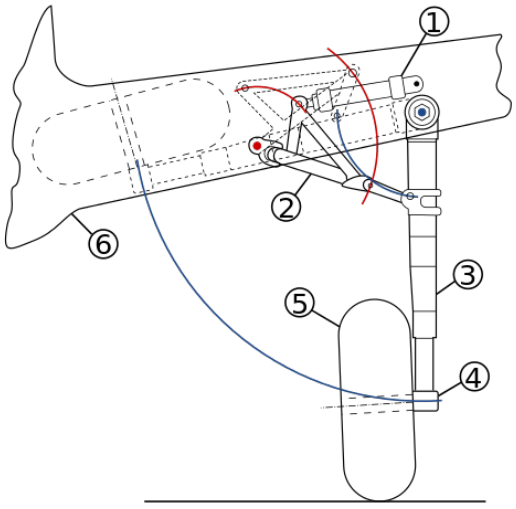


Figure 9: Our design choice [4].

3.2 Dimensions

3.2.1 Initial Dimensions

After finalizing the design, the next challenge we encountered was determining the dimensions. We were uncertain about the appropriate size for the model and the truss dimensions required for our retractable landing gear mechanism.

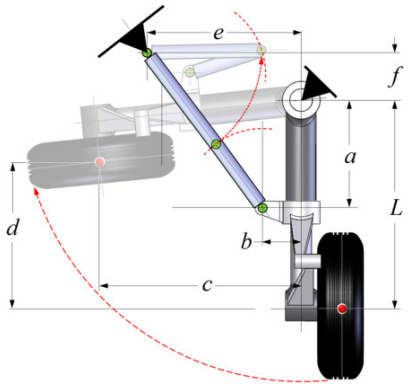


Figure 10: Diagram representing dimensions of the retractable landing gear [6].

As shown in Figure 10, designing a functional retractable landing gear mechanism requires numerous dimensional parameters. Some of these values need to be calculated, while others must be estimated based on engineering judgment and design constraints.

For our preliminary research, we analyzed the retractable landing gear mechanism of the Boeing 737, specifically focusing on the main landing gear system.

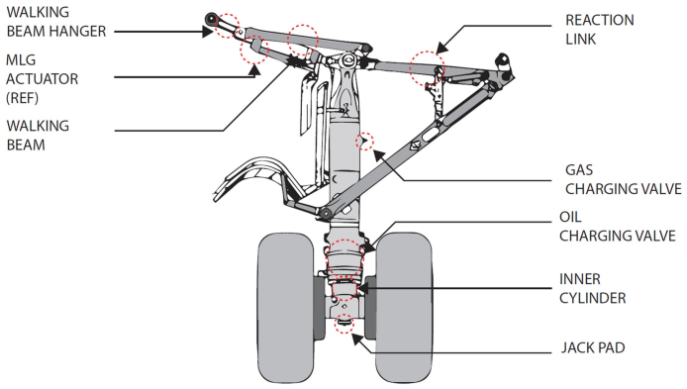


Figure 11: Retraction system of a Boeing 737 [12].

Using the Boeing 737 as a reference, we aimed to develop an alternative system based on our design. By analyzing the aspect ratios from the image, we were able to extract key values such as a and e. Additionally, we considered the dihedral angle of the Boeing 737's wing, which is approximately 6° [1]. As a result, we positioned the pivot point, represented by dimensions e and f, lower than that of the Boeing 737 to account for the dihedral angle's effect on the retraction mechanism.

Ultimately, we arrived at the following preliminary design.

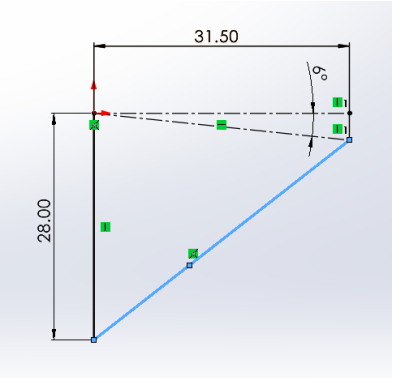
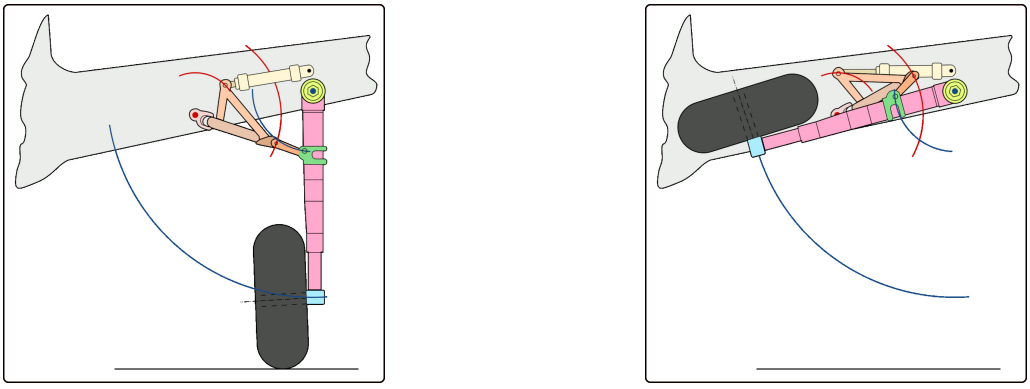


Figure 12: Preliminary retractable landing gear dimensions. Blue dimensions are yet to be determined. Done in Solidworks by us.

For the size of our product, we decided to use a 1:1 cm scale after comparing it with another model of a retractable landing gear. We wanted our mechanism to be clearly visible while also being able to withstand a proper human load, ensuring that the durability of the system could be demonstrated effectively.

3.2.2 Finding the Remaining Dimensions

Determining the final two dimensions of the trusses is not straightforward. We began by listing the key conditions that would guide our optimization process. Firstly, the mechanism must fully extend to align with the 6° dihedral angle of the wing so that the landing gear integrates seamlessly with the fuselage and wing structure. Secondly, the trusses must be designed to avoid any collisions during the retraction and deployment process. Finally, the entire mechanism must remain compact, ensuring that no part of the trusses extends beyond the wing's surface when retracted. Our goal was to achieve a configuration similar to the one shown in the images below.



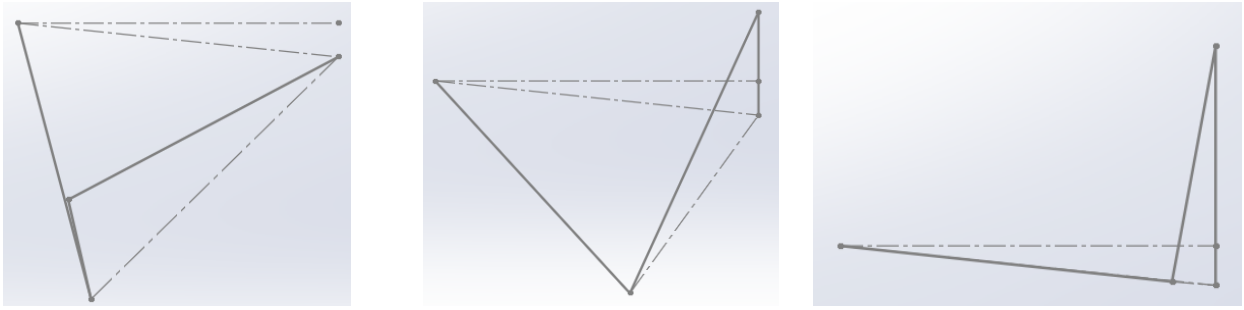
(a) Landing gear in the deployed (open) state. **(b)** Landing gear in the retracted (closed) state.

Figure 13: Comparison of open and closed states of a properly dimensioned landing gear [10].

The position of the joint connecting the two trusses is critical, as its motion drives the entire landing gear mechanism. In the fully extended state, the trusses must be collinear with each other and aligned with the side pivot point to ensure proper locking of the mechanism. This aspect will be discussed in more detail in the locking mechanism section.

Poorly chosen dimensions can lead to various problems, such as the landing gear failing to retract due to truss collisions, incomplete deployment (underextension), or parts of the trusses

extending beyond the wing, making the design uncompact. These potential issues are illustrated below.

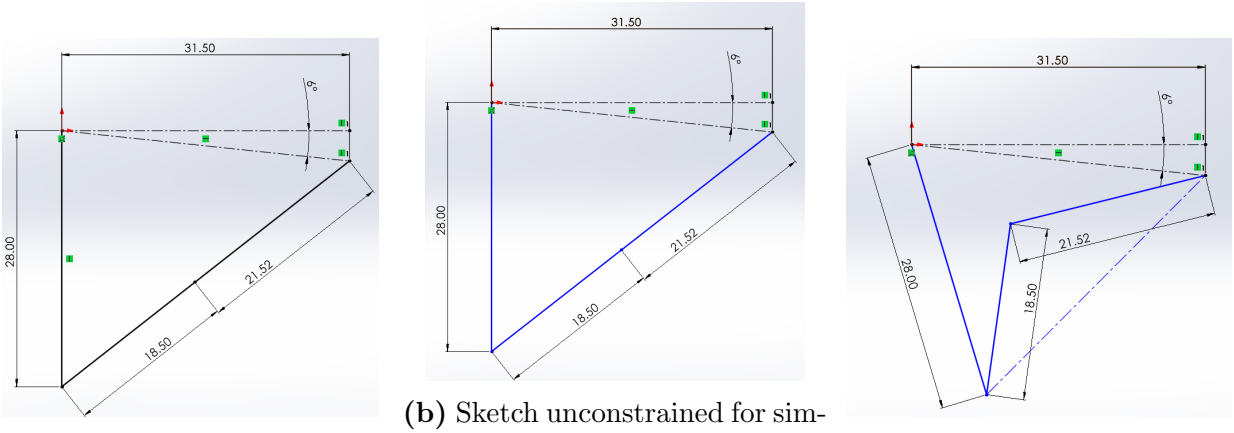


(a) Truss collision. **(b)** Truss too short. **(c)** Uncompact design.

Figure 14: Possible problems caused by incorrect truss dimensions.

To determine the appropriate truss dimensions, we were unable to find any direct formulas or guidelines. While we could have simply used the dimensions from the Boeing 737, we decided to derive them ourselves. We devised two possible methods for calculating the values.

The first method, which we used, involved creating a parametric sketch in SolidWorks. We input our initial known dimensions and left the unknown dimensions as free parameters. By iteratively adjusting these values and allowing some joints to remain unconstrained, we were able to simulate the movement of our landing gear mechanism. This process enabled us to visually assess whether the system met our design constraints. We aimed to find dimensions that allowed the retraction system to function smoothly and continuously while adhering to our optimization criteria. The iterative process in SolidWorks is illustrated below.



(a) Input of initial values. **(b)** Sketch unconstrained for simulation. **(c)** Motion analysis.

Figure 15: Step-by-step process for iterating and refining truss dimensions in SolidWorks.

After several iterations and refinements, we finalized the following truss dimensions.

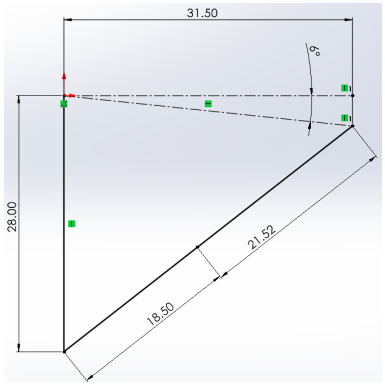


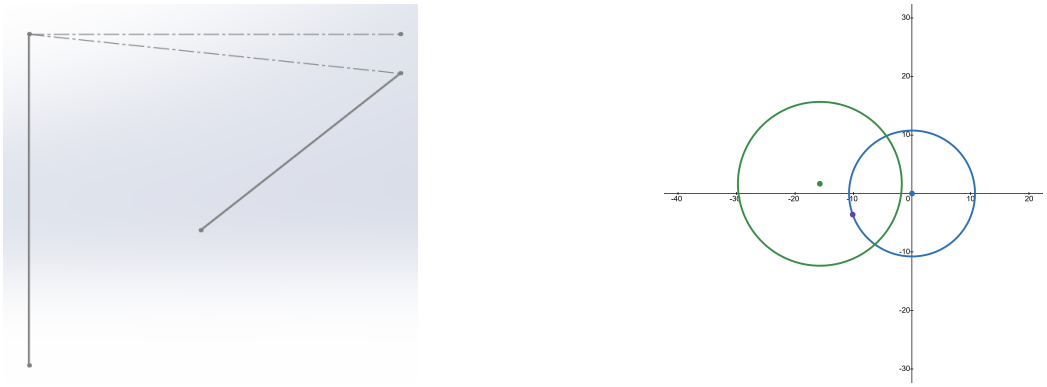
Figure 16: Final optimized truss dimensions.

Our second approach, which we conceptualized but did not fully implement, involved a mathematical calculation method. This method relies on determining the minimum and maximum possible values for the trusses.

Since our model has two pivot points, each connected to a truss, the movement of these trusses follows a circular trajectory around their respective pivot points. Given that their lengths remain constant, their positions can be described using circular motion equations. Using the vector representation

$$\begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

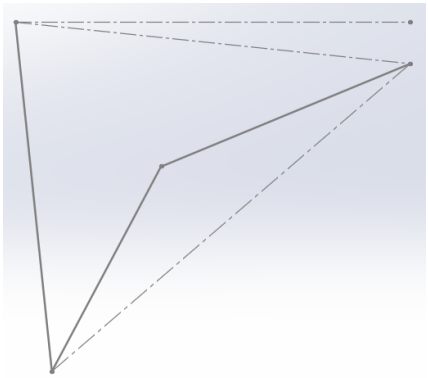
we can define the position of a point rotating around a fixed pivot. This is depicted in the images below.



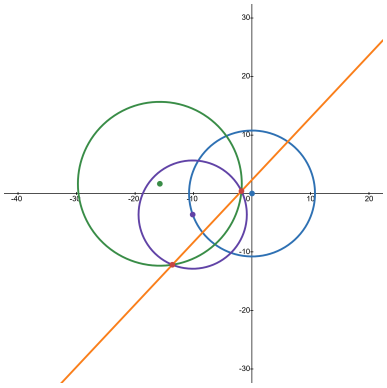
(a) Circular motion representation in SolidWorks. **(b)** Mathematical visualization in Desmos.

Figure 17: Illustration of circular motion used to determine truss positioning.

From this, we deduced that the second truss would also follow a circular path around its own pivot point, forming a second circle. The intersection of these two circles determines the final attachment point of the truss to the landing gear beam. This geometric approach is visualized below.



(a) Intersection of circles in SolidWorks.



(b) Intersection of circles in Desmos.

Figure 18: Geometric approach to determining the final attachment point of the truss.

In this method, the radii of the circles represent the truss dimensions. By converting these points into vectors as functions of the angle θ (the angular displacement of the first truss), we could establish constraints ensuring that the trusses do not collide with the landing gear beam. This would provide a range of valid values for the mechanism to function properly. Finally, we could optimize the dimensions by selecting the smallest valid value for the first truss, ensuring the most compact and efficient design.

Although we developed this theoretical method, we were unable to fully translate our mathematical understanding into concrete equations at the time. However, we plan to revisit and complete this approach in the future.

3.3 Diagrams

3.3.1 Kinematic

The retractable landing gear mechanism involves motion, meaning a kinematic diagram can be created to analyze its behavior.

This mechanism contains three fixed pivot points: one that directly supports the wheel beam, one that manages side forces to provide stability and enable gear retraction, and a third that serves as the mounting point for the actuator. The kinematic diagram is shown below.

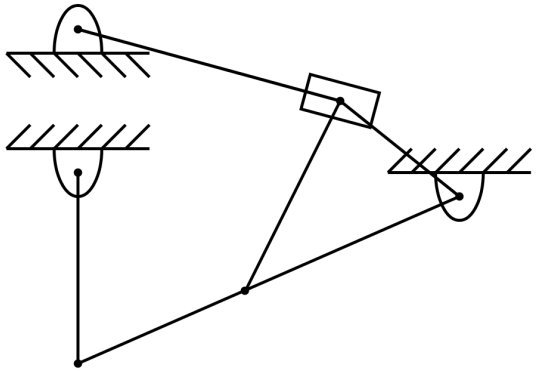


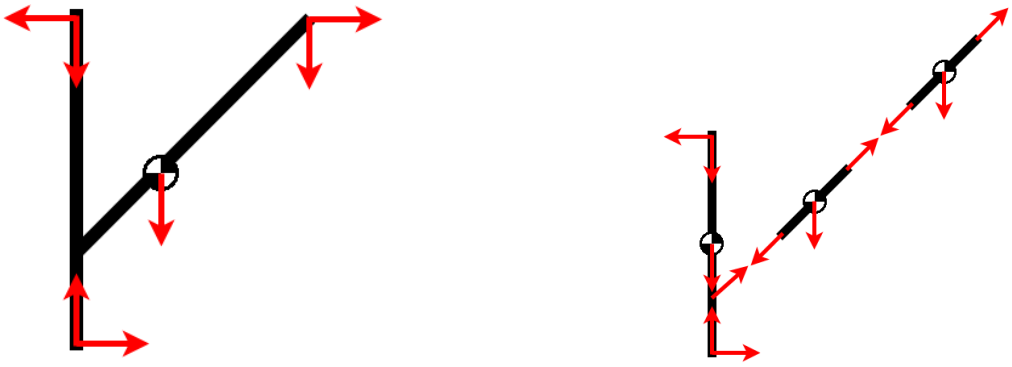
Figure 19: Kinematic diagram of the landing gear in the extended position.

As seen in the diagram, the system consists of the wheel beam and two trusses (#1 and #2), which are connected through two pin joints. The motion of the wheel beam and truss #2 follows a circular path around their respective fixed pivot points, while truss #1 rotates around its pin connection.

Additionally, the actuator operates in a linear direction and is connected to a rotating triangular linkage. As the actuator extends or retracts, it generates a circular motion in the triangle, which, in turn, applies force to the pin where all three trusses connect. This interaction drives the rotation of the entire landing gear retraction mechanism.

3.3.2 Free-Body

The landing gear on its extended position is designed to support all forces exerted on the wheel on the moment of landing, so let's analyse all forces on the landing gear system in this position on the following free body diagram.



(a) Full **(b)** Members.

Figure 20: Free body diagrams on the extended position.

List of forces

1. The forces exerted on the wheels are, firstly a sideway force caused by misalignments on the landing or crosswinds, secondly a normal force that supports the weight of the whole airplane, and thirdly a braking force.

2. Weight of the landing gear system, i.e. main shaft, wheel and trusses.

3. Both this points are pin connections that fix the landing gear system to the airplane and that support the weight of the airplane and the reaction forces from the landing.

On the Figure 20b we can observe that, excluding the weight of the member itself which is negligible compared to the other forces on them, member 1 and 2 act as two force members, and therefore no forces besides the weight of the triangular truss are exerted on it, and by consequence no force is applied on the actuator on the extended position.

3.4 Locking Mechanism

3.4.1 Forces

The locking mechanism plays a crucial role in aircraft operations, including holding, taxiing, and landing. It prevents the landing gear from retracting unintentionally due to external forces or its own weight. There are two primary methods of locking.

The first method relies on a fundamental understanding of force equilibrium. As previously mentioned, the two trusses were deliberately designed to be collinear in the fully extended position, forming a two-force member, assuming that weight is negligible in load transfer. As a result, no acceleration occurs at the connection point. This behavior is also evident in the Free-Body Diagram, where forces at the pin connection cancel each other out.

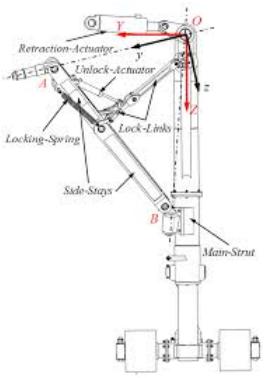


Figure 21: Free body diagram representing forces acting on the pin.

Conversely, if the trusses were not designed to align in this manner, acceleration would be introduced at the joint. This effect can be calculated, and the results are illustrated in Figures **??** and **??**.

3.4.2 Precautionary Additional Mechanism

Although the two-force member mechanism is expected to provide a reliable lock, a precautionary mechanism is necessary to account for potential factors such as metal deformation, high-impact forces, and scenarios where the weight cannot be considered negligible.

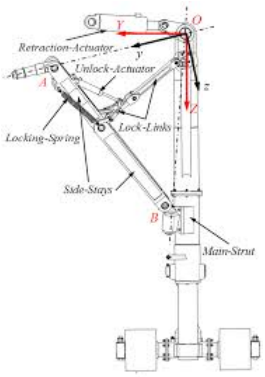


Figure 22: Diagram showing additional locking mechanism [13].

It's disadvantage is that it requires some actuator or force to release the lock, so we decided to go only with the 2 force member solution.

3.5 Actuator

The actuator is a key part of the system, since it is the source of motion of the landing gear, therefore a reliable and functional, yet easily manufacturable and low-cost actuator design had to be done.

3.5.1 Design choices

Given the means of production and materials available, and given the absence of an actual motor, we quickly abandoned the possibility of hydraulics, pneumatics and electric systems, thus holding on to two mechanicals systems, rack and pinion or a bolt-and-nut-like system. Bearing in mind that it would need to support axial load, we opted for the latter.

With a system in mind, it came time to designing.

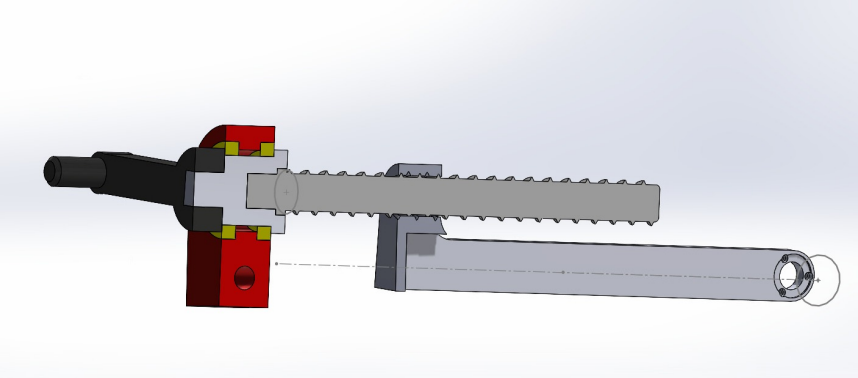


Figure 23: CAD model of actuator's assembly.

Our design involves a part fixed to the frame by a pin connection that supports the screw while allowing its rotation, the screw itself with a crank, and part with an inside thread that will be connected to the trusses and that will translate the rotational movement of the screw to a linear movement, as long as this part itself cannot rotate.

The first models involved a mechanism with a drive to stop the rotation without putting any unwanted loads on the trusses, however, for the sake of simplification and costs, that idea was dropped, instead the current design was adopted, where the rotation is stopped by the triangle truss, which is much simpler, even though it does create unwanted loads on the system.

The screw was broken down in 3 parts, a threaded shaft, a middle part that would be in contact with ball bearings, and the crank. They were all connected using square spindles since this is easily manufacturable with a 3d printing machine and yet extremely reliable and resistant. The assembly of the whole actuator was quite straightforward, since precise gaps were designed for the ball bearings and the crank acted as fastening for the whole system.

3.5.2 Tests

The inside and outside threads are crucial to ensure a smooth operation of the actuator, they are, essentially, what translates the rotational motion to linear, therefore, once the pitch of the threads were decided to be 9,5mm, which is about 5% of the total extension length, so about 20 full rotations for a full extension or retraction, we moved on to conducting tests with sample sizes threads of different gaps between the diameter of the screw and that of the threaded through-hole in order to achieve smoothly working threads. These tests, about 7 different sizes, cost 4.5 euros and 30g of PLA until the final dimensions were reached.

3.6 Process

3.6.1 Material selection

Landing gear supports the aircraft's weight while it is on the ground and while it is landing, therefore the pressure and the forces acting on it can be quite high. In order to cope with this environment, the materials for a landing gear must be able to withstand high loads of impact forces and other environmental stress.
Primary materials used in a landing gear are: high-strength steel alloy, which has very high strength and fatigue resistance, titanium alloy (Ti-6Al-4V), a strong and light material, it is lighter than steel. Other materials consist of: steel alloy, titanium, maraging steel, carbon composite, aluminum alloy. All of these materials are carefully selected and used in different parts where the working environment of the part requires a different approach.

For our own design however, we had 2 different materials to choose from, wood and plastic (PLA). We wanted to make our shafts parts mostly from wood to have stability and high-strength, and for our main shaft or "shock strut", we used a steel shaft to better resemble the weight of the gear. Other small parts such as washers or bushes had to be given to us by our tutors so we didn't have many choices. That was our original planning, but after a few days of reconsidering and redesigning we realised that we wanted to have a detailed and good looking product, and making most of our shafts out of wood is not the viable option since you cannot add details to wood and carving them to our design is nearly impossible, so we used PLA for most of our parts, this material while is more brittle and not as stable as wood, it gives a very professional look to our product and by printing our parts, we can have details that we wouldn't have on wood. Overall we made an excellent choice choosing PLA over wood, here is our table of what parts we have used and what materials were they made of.

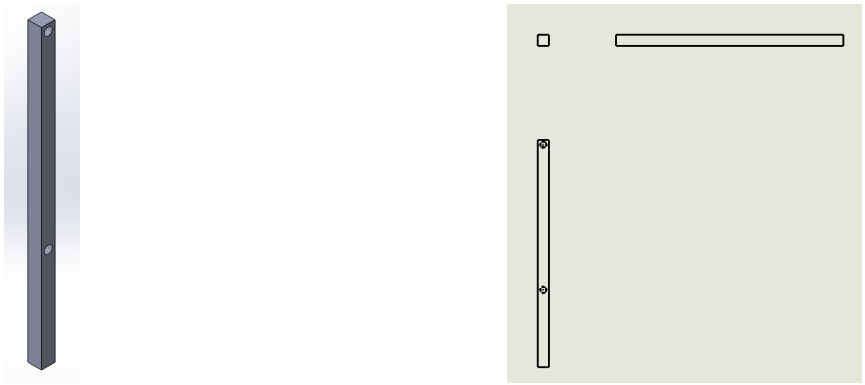
Name	Amount	Material
40mm elcoms	8	Steel
30mm elcoms	3	Steel
20mm elcoms	4	Steel
50mm elcoms	1	Steel
L connections	12	Steel
Straight connections	8	Steel
10mm small shaft	6	Steel
Bush	4	Rubber
Small caps	10	PLA
Actuator	1	PLA
Main triangle	1	PLA
Connecting beam	1	PLA
Frame beams	12	Steel

Table 1: List of items with their amounts and materials.

Our product is not yet complete since in the manufacturing process there was a lack of materials and parts we could acquire, there was also a shortage of bushes and connections which led us to have to buy some ourselves, the printing machines were also too crowded so we had to print all our parts in a different building.

3.6.2 CAD Model

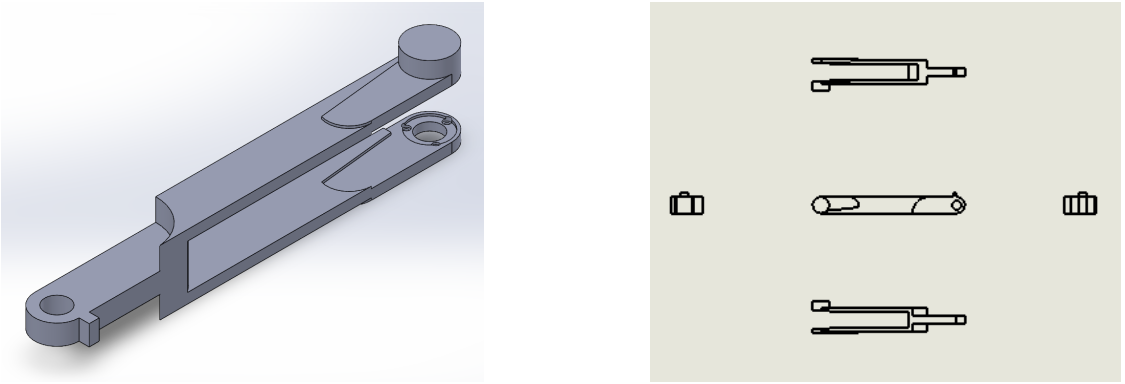
Wheel Beam The wheel beam is a straight structural component extending from a pivot point down to the ground. It features two through-holes designed to accommodate bushings for holding a pin securely. We specifically chose an Elcom profile for this part, making the design simple and efficient.



(a) 3D View **(b)** Technical Drawing

Figure 24: Visualization of the wheel beam.

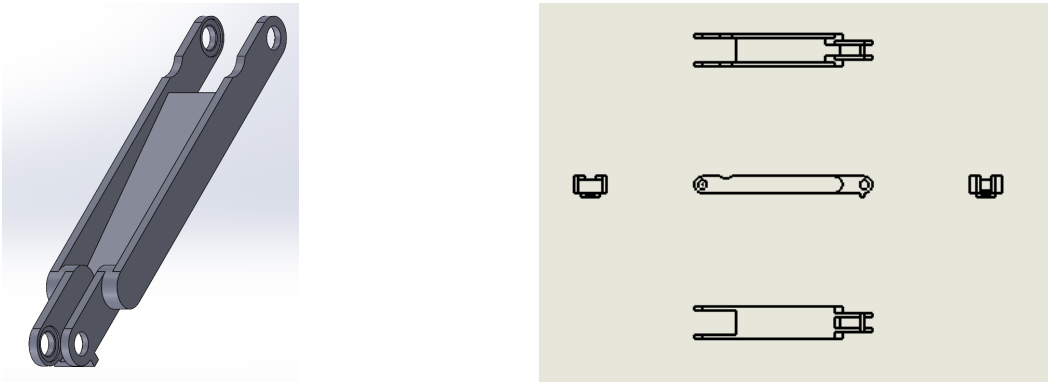
Truss #1 Connects the wheel beam to truss #2 through bushings and pins. It features an integrated cap and an external cap to securely lock the pin inside the part. The inner cut is specifically designed to prevent collisions between the wheel beam and the material during rotation. The outer fillet aligns with truss #2 and the triangular component in the retracted state. Additionally, an opposite-direction lock prevents unintended movement when the mechanism is opening.



(a) 3D View **(b)** Technical Drawing

Figure 25: Visualization of Truss #1.

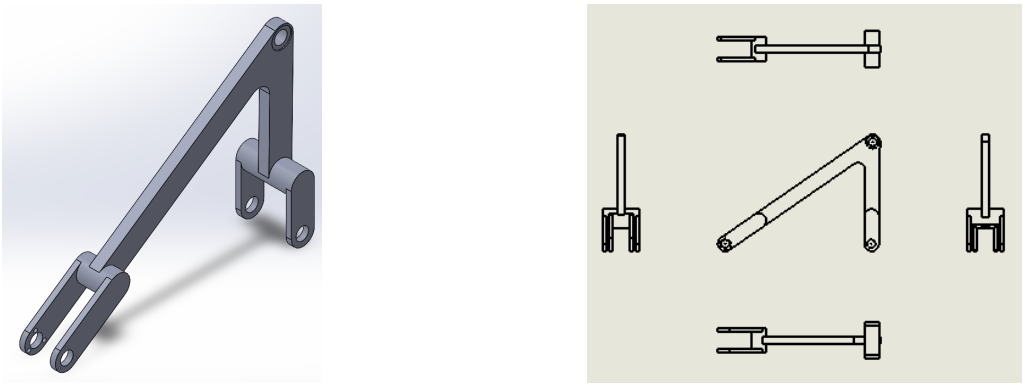
Truss #2 Connects to truss #1 and the fixed pivot point, with the triangular component mounted on top. Like the other parts, it has bushing holes for shafts. Structurally, it consists of two mirrored sections joined together. The small opening accommodates truss #1, while the larger opening allows space for the wheel beam in the retracted state. The inclined flat surface inside acts as a support bed for truss #1 when fully retracted. Similar to truss #1, it also includes a locking feature to prevent unintended rotation when fully extended.



(a) 3D View **(b)** Technical Drawing

Figure 26: Visualization of Truss #2.

Triangle The triangular component features multiple bores for bushings and openings for trusses #1, #2, and the wheel beam. Due to its size, the legs were added separately. An adapter at the top allows attachment to the actuator, facilitating controlled movement of the mechanism.



(a) 3D View **(b)** Technical Drawing

Figure 27: Visualization of the triangular component.

Full Model The full assembly includes all components required for the landing gear mechanism. The complete list of parts is as follows:

- **Main structural components**: Wheel beam, truss #1, truss #2, triangle.
- **Mechanical connections**: Washers, shafts, pins (small shafts), bushings.
- **Actuation system**: Actuator, crank, actuator adapter, actuator holder.
- **Supporting frame**: Elcom frame.

3.6.3 Manufacturing

The manufacturing process of the landing gear model focused on using accessible materials and 3D printing technology. The key structural components—wheel, trusses, and actuator—were



(a) Trusses Assembly **(b)** Complete Assembly

Figure 28: Visualization of the full landing gear assembly.

designed in CAD software and printed using PLA. Bushings and pins were incorporated to ensure smooth movement and durability.

We connected the different parts of the trusses using shafts and then connected these to the wheel beam (50 cm ELCOM) using pins we then attached the actuator to the rest of the structure using shafts. Finally we connected our final structure to the frame using shafts and pivot points.

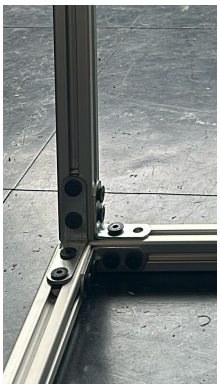
We used ELCOM aluminum profiles to manufacture the overall frame that would support our structure. For the legs of the frame, we used 4x60 (40+20) cm ELCOM that would help the rectangular base with a width of 0 cm. For the connections of the different aluminum profiles, we had to buy straight (Figure 29a) and L-shaped (Figure 29b) stainless steel hole plate connectors from a hardware store (Etablissements J Thollot et Fils) since there weren't enough provided for us by the GM Factory. This was an inconvenience for us as it made us lose a lot of time and pay out-of-pocket. However, the upside of using these connections instead of 3-D printed ones is that our frame was very sturdy and could handle the rest of the structure on it. We tested out 2 3-D printed connections that ended up snapping after tightening the screws in. All in all, for the frame, we used 8x40, 3x30, 4x20 ELCOM.



(a) Trusses Assembly



(b) Complete Assembly



(c) Complete Assembly

Figure 29: Visualization of the full landing gear assembly.

4 Conclusion

4.1 Final Product

For the final product of our landing gear, we were able to get a fully functioning mechanical model that would retract and extend with the respective application to the actuator. Our model, due to all the difficulties and setbacks we had throughout the design process, was done at 7 p.m. the night before the presentation which did not give us enough time to test its functionality. However, due to the fact that we thoroughly studied everything theoretically, we had no issues at all after the final assembly.

Our final landing gear is a scaled-down, fully mechanical retractable landing gear model that efficiently shows the key principles of landing gear design, including self-locking mechanisms, retraction systems, and load-bearing trusses. The use of a manually operated actuator allowed for a simplified yet functional representation of real aircraft landing gear without reliance on hydraulic or electrical components.

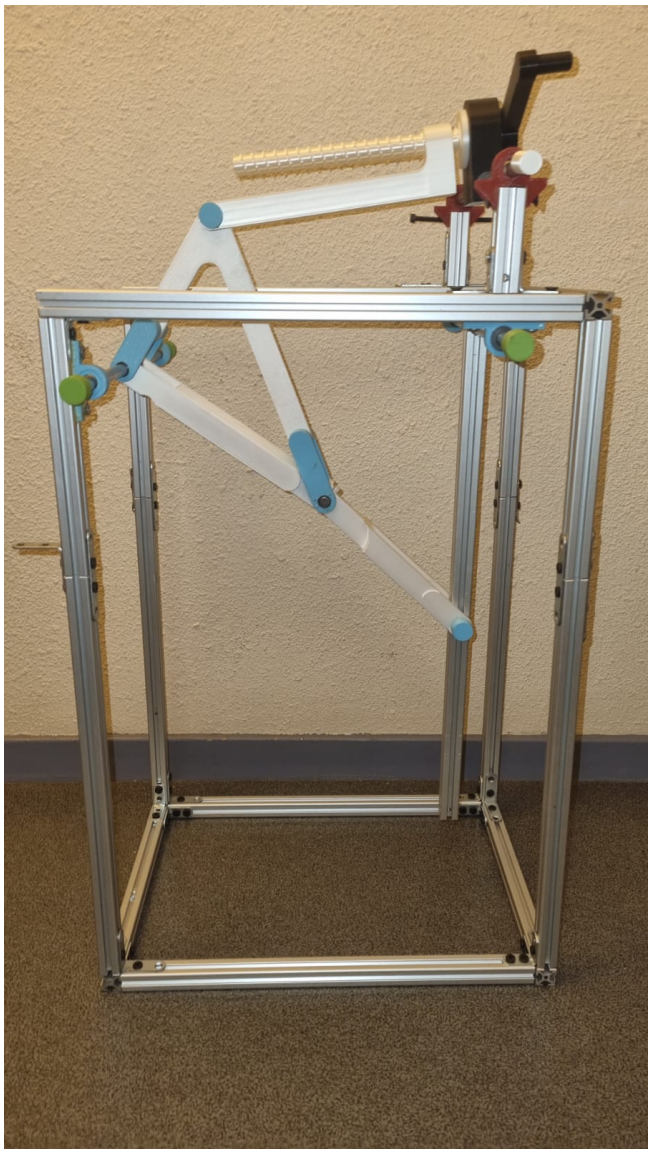


Figure 30: Final product.

4.2 Possible Improvements

While the model efficiently fulfills its intended purpose, several enhancements could significantly improve its performance and realism. One key improvement could be the refinement of the locking mechanism to account for material deformation over time, ensuring consistent stability and preventing unintended retraction due to wear or external forces. Additionally, integrating a motorized actuator would allow for automated operation, making the system more representative of real aircraft landing gear, which relies on hydraulic or electric actuators. Upgrading to stronger structural materials, such as aluminum alloys or composite materials, could better simulate the stress conditions experienced in actual aircraft and enhance durability. Furthermore, advanced mathematical modeling and simulations could optimize the truss dimensions and actuator efficiency, ensuring smoother retraction and extension while minimizing unnecessary weight and bulk. These refinements would make the model not only more functional but also a more accurate representation of full-scale landing gear systems.

Moreover, in terms of project organization, we would want to improve the availability and accessibility to the 3-D printers. All of our reserved times for the printers were not respected by other teams and were not taken into account by anyone. We would also seek to improve the communication with other members that were involved in aiding with the project (specifically GM factory) as most of the parts we needed were not available. As a conclusion, the process was not easy to say the least, but our team endured and worked hard together to get a successful result of our landing gear project.

Finally we also thought about doing an emergency remove mechanism, however getting a right thread and locking mechanism would be an project on its own.

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5 Appendix

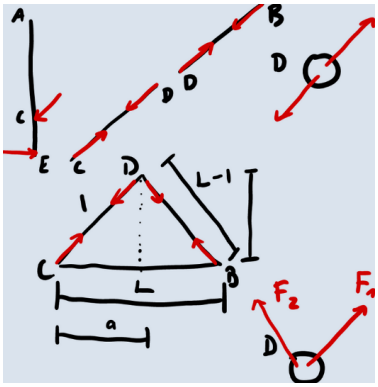


Figure 31: Acceleration at pin calculation

$$F_x = 0,\ W = 0 \tag{1}$$

$$a = l\cos\alpha,\ b = l\sin\alpha \tag{2}$$

$$\cos\theta = \frac{L-a}{L-l} = \frac{L-l\cos\alpha}{L-l} \tag{3}$$

$$\sin\theta = \frac{b}{L-l} = \frac{l\sin\alpha}{L-l} \tag{4}$$

$$x:\ 0 = F_1\cos\alpha - F_2\cos\theta = F_1\cos\alpha - F_2\frac{L-l\cos\alpha}{L-l} \tag{5}$$

$$\therefore\ F_1 = F_2\frac{L-l\cos\alpha}{(L-l)\cos\alpha} \tag{6}$$

$$y:\ F_1\sin\alpha + F_2\sin\theta = F_2\frac{L-l\cos\alpha}{(L-l)\cos\alpha}\sin\alpha + F_2\frac{l\sin\alpha}{L-l} = F_2\frac{L\tan\alpha}{L-l} \tag{7}$$

Figure 32: Acceleration at pin calculation